

Car Price Prediction Analysis

A Comparative Study of Linear Regression and Random Forest Regression

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COURSE

Data Science & Machine Learning Bootcamp

1. Project Implementation & Model Training

What Was Implemented?

I built a complete machine learning pipeline inside a Jupyter Notebook to predict used car prices. The project utilized the structured, cleaned dataset generated during **Assignment Three**.

Training Methodology

- **Linear Regression Model**
 - Trained as a baseline parametric model.
 - Mapped linear equations between continuous features (e.g., odometer mileage, engine capacity, manufacture year) and the target vehicle price.
- **Random Forest Regressor**
 - Implemented as an ensemble tree-based algorithm.
 - Trained using multiple parallel decision trees to capture non-linear degradation curves and multi-feature interactions.
- **Data Split:** Both models were trained and evaluated using an **80/20 train-test split** framework.

2. Comparison of Models & Sanity Check

Sanity Check Observations

During model validation and manual sanity checks (testing specific car profiles, such as an old high-mileage car versus a new luxury car), the predictions deviated heavily:

- **Linear Regression Deficiencies:** Predicted unrealistic values for older, high-mileage vehicles. Because it operates on a strict straight-line function, it occasionally projected prices that dipped into negative ranges or near-zero, which does not align with commercial automotive markets.
- **Random Forest Realism:** Provided highly stable, realistic, and strictly bounded price estimations for identical edge-case vehicles.

Why Random Forest Gave More Realistic Results

- **Non-Linear Thresholds:** Vehicle depreciation slows down over time and levels off at a baseline structural value. Linear Regression assumes a continuous, infinite rate of price decline.
- **Step-Function Processing:** Random Forest captures categorical interactions (e.g., brand prestige combined with mechanical wear) via logical splits, making its real-world pricing behavior far superior.

3. Understanding Random Forest

Definition and Mechanism

In my own words, **Random Forest** is an ensemble learning algorithm that constructs a collection (a "forest") of hundreds of individual **Decision Trees** during the training cycle.

Core Operational Workflow

- **Bootstrapping (Random Selection):** Each individual tree is trained on a distinct, randomly sampled subset of the training data and a random selection of input features. This injects mathematical diversity.
- **Independent Predictions:** When a new car's features are evaluated, every single tree outputs its own independent price estimate.
- **Averaging Output:** The algorithm aggregates all individual tree calculations and delivers the final prediction by averaging the scores. This group consensus controls variance and effectively protects the model against overfitting.

4. Metrics Discussion & Evaluation

Empirical testing on the unseen validation dataset revealed a stark contrast in statistical performance metrics:

PERFORMANCE LEADER

The **Random Forest Regressor** outperformed Linear Regression across all operational parameters, yielding a significantly higher Coefficient of Determination (R^2) alongside drastically reduced error scales (**MAE** and **RMSE**).

Structural Strengths & Weaknesses

Model	Strengths	Weaknesses
Linear Regression	Exceptional training speed, computationally lightweight, and mathematically transparent (high interpretability).	High Bias (Underfitting). Fails to map non-linear trends or identify how specific thresholds (e.g., passing 100,000 km) impact consumer buying psychology.
Random Forest	Low Bias and Low Variance. High predictive accuracy, naturally handles mixed feature types, and isolates dataset noise.	Operates largely as a "Black Box" model, making feature weights harder to explain linearly; cannot extrapolate price vectors beyond its training boundaries.

5. Final Findings & Core Preference

Model Preference

Based on the empirical evidence and performance metrics derived from this project, I firmly prefer the **Random Forest Regressor** for deploying real-world vehicle price prediction engines.

Final Justification

Automobile asset evaluation is complex and heavily affected by non-linear factors such as seasonal market changes, regional preferences, and trim levels. Random Forest processes these intricate feature correlations naturally without requiring exhaustive polynomial engineering. While Linear Regression remains an invaluable tool for establishing rapid data baselines and checking initial trend vectors, it lacks the technical depth required to handle real-world economic variances. Random Forest consistently generates reliable, mathematically stable, and accurate price distributions, making it the ideal framework for modern automotive intelligence systems.